

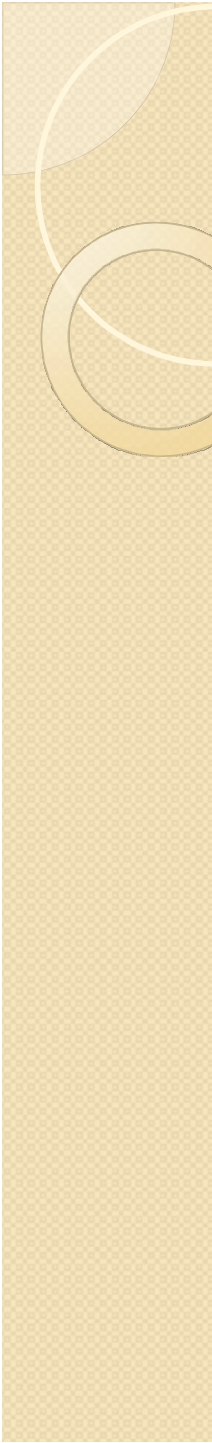
Tech-Edge Lab

Topic: -

**Generative AI & Intelligent Application
Development Bootcamp**

Sub-topic:-

**Industry-Oriented Training Program on
LLMs, AI Tools, AI Agents & Vibe Coding**

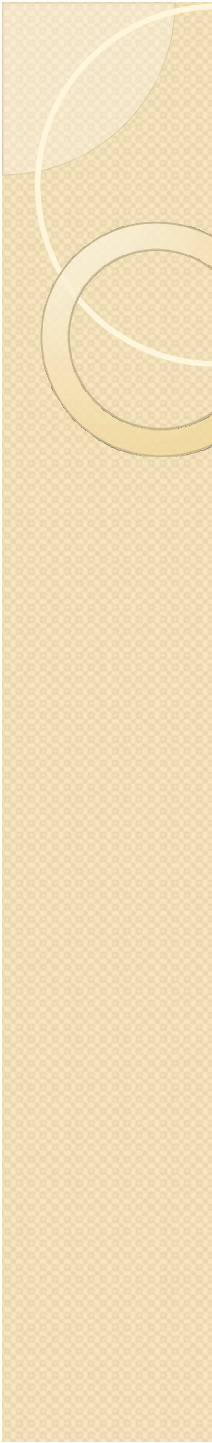


Module 2:- AI Foundations

Lecture 3:- How AI works(training, inference, datasets, parameters, tokens, context window, chatgpt token prediction)

Index

S.No.	Topic
1.	Human learning Vs AI learning
2.	Data, Datasets, training
3.	Parameters
4.	Small Model Vs Large Model
5.	Inference
6.	Token, token prediction
7.	Context window
8.	Complete AI workflow
9.	Class Challenge
10.	Industry Oriented patterns
11.	Activity



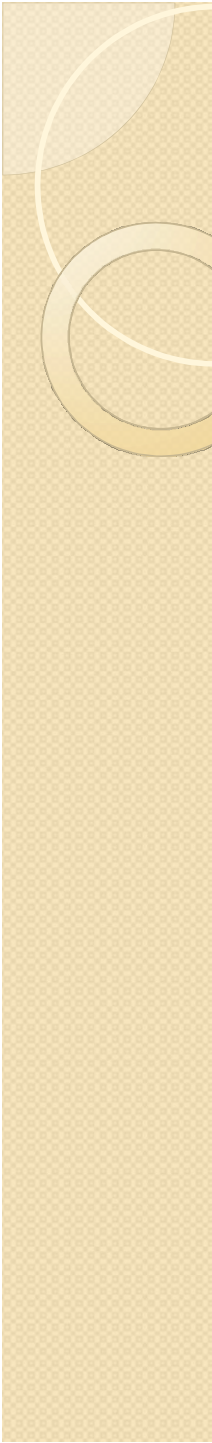
Recap

Yesterday we learned

- AI
- ML
- DL
- NLP
- Computer Vision
- Robotics
- Generative AI

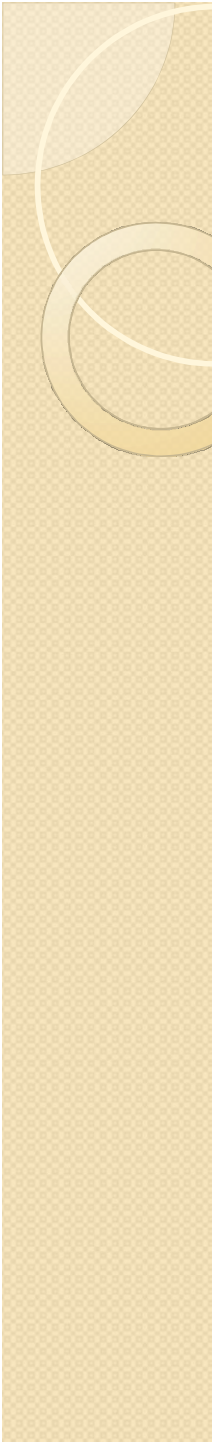
We know ChatGPT is AI...

But how does ChatGPT actually learn?



Activity 1

Q- How did you learn to ride a bicycle?



Answers:-

- Practice
- Mistakes
- Parents
- Experience

Now connect

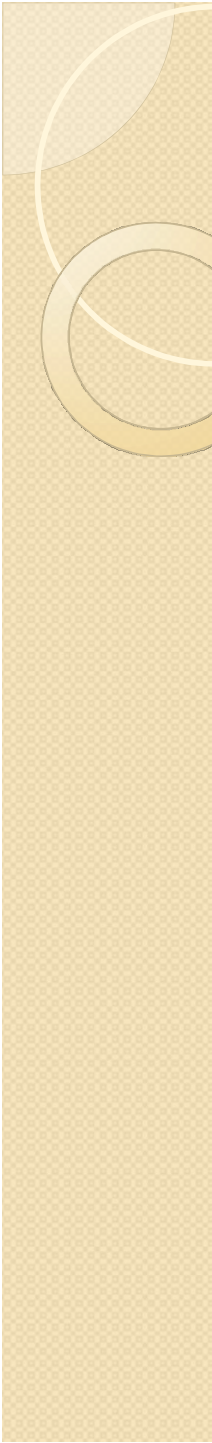
Humans learn from experience.

AI learns from data.

Human Learning vs AI Learning

Human	AI
Books	Dataset
Practice	Training
Memory	Parameters
Thinking	Inference
Speaking	Generated Output

Every AI model goes through a learning process just like students preparing for exams.



What is Data?

Examples

Images

Books

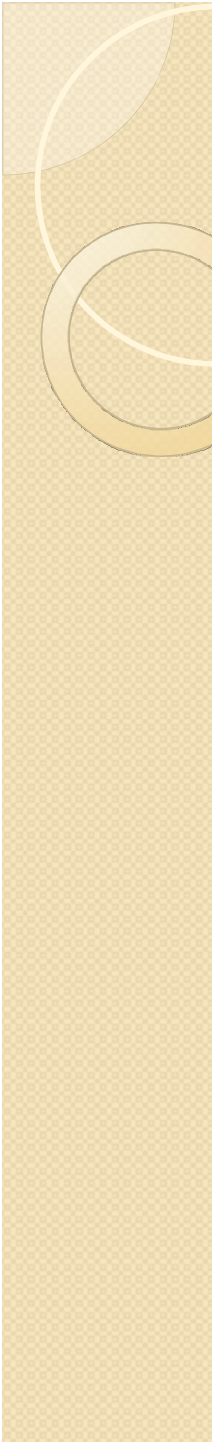
Websites

Videos

Audio

Code

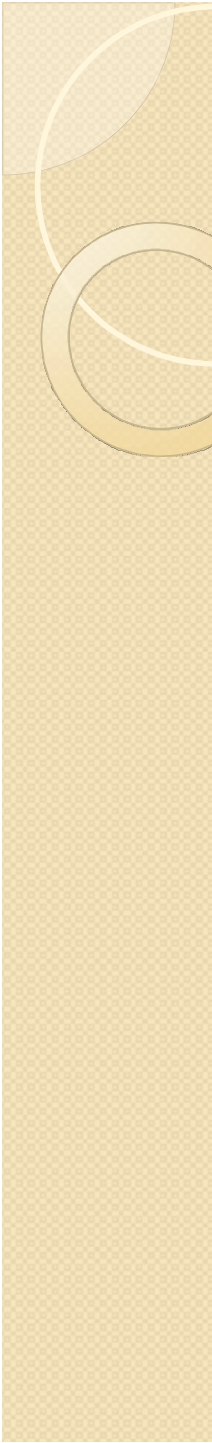
**Without data,
AI cannot learn.**



Activity

Q- Can AI learn if no data exists?

A- Yes/No



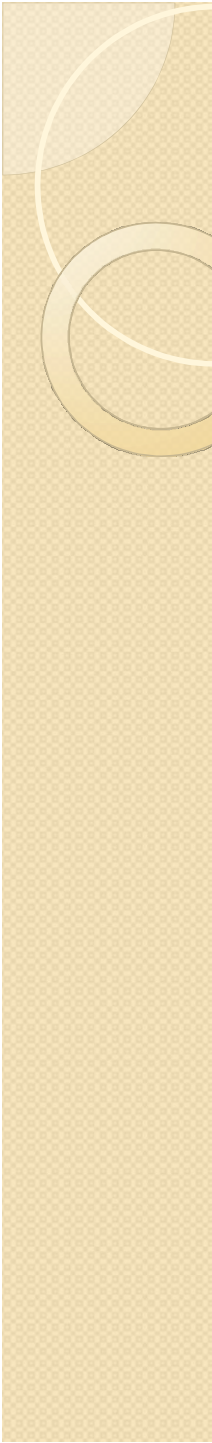
Dataset

Definition

A dataset is a collection of organized data used to train an AI model.

Examples

- Millions of books
- Wikipedia
- Research papers
- Source code
- Images
- Conversations



Activity 2

Q- Think 20 images of cats and dogs (or simple examples).

Q- How will AI distinguish cats from dogs?

Look at these 20 images of Cats and Dogs



QUESTION

How will AI distinguish cats from dogs?



STUDENTS ANSWER



By seeing many examples!

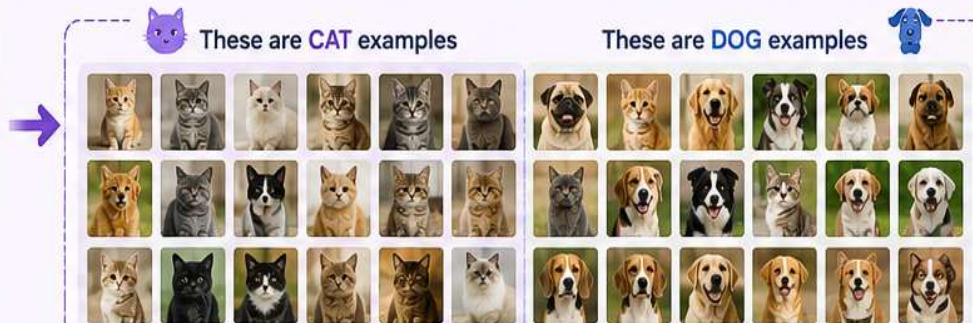


Introducing: DATASET

A **Dataset** is a large collection of examples (data) used to teach AI.



Many images (labeled examples)



AI learns patterns from the dataset and can then predict new images!

After Learning...



Cat



Dog



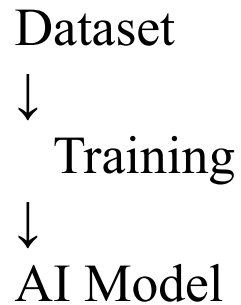
The more and better the dataset, the smarter the AI!



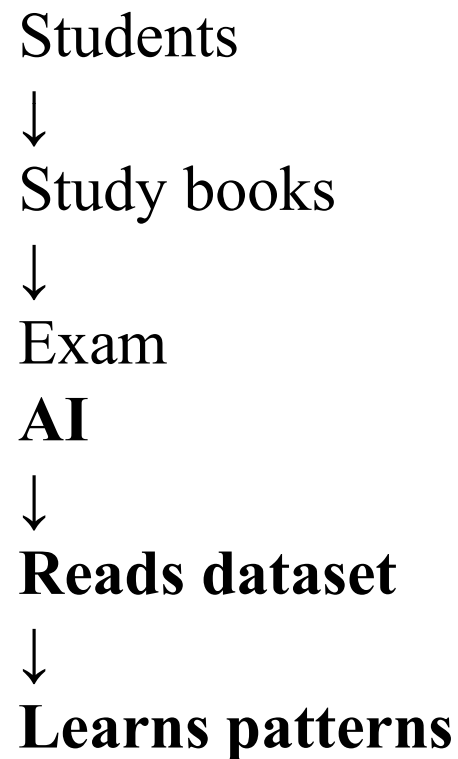
AI doesn't magically know the answer. It **LEARNS** from **DATA**!

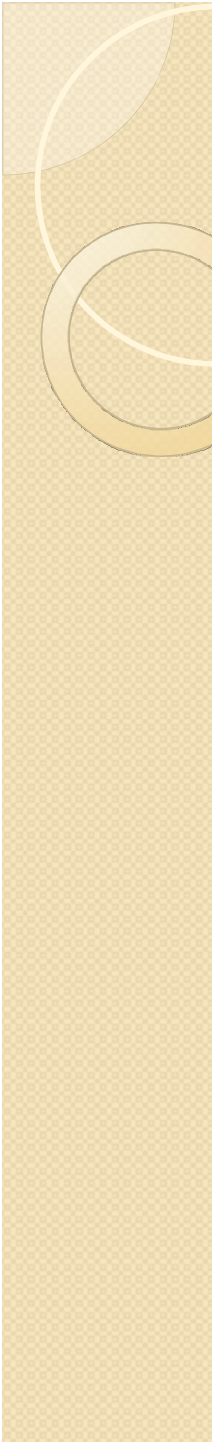
Training

Training is the process where AI learns patterns from data.



Example:-





Activity

Q- How many times do students revise before exams?

A-

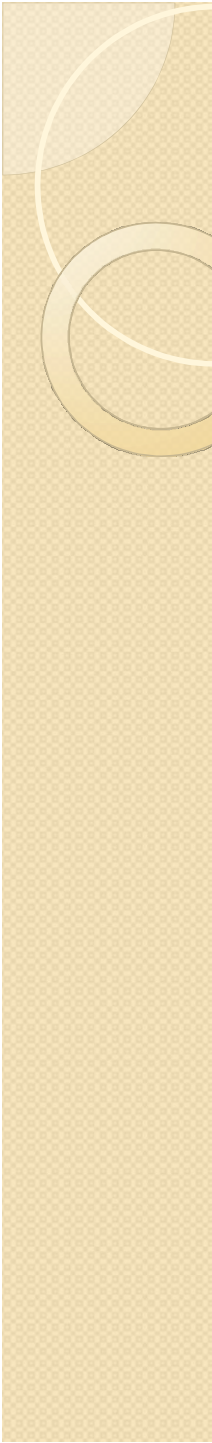


“AI also repeats training many times.”

Real Example:-

ChatGPT was trained on

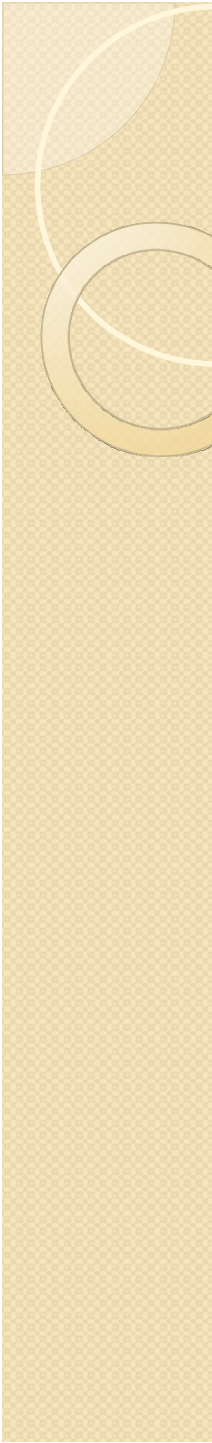
- Books
- Websites
- Code
- Articles
- Research Papers
- Millions/Billions of examples.



Discussion

Q- Why does ChatGPT know Java?

A-

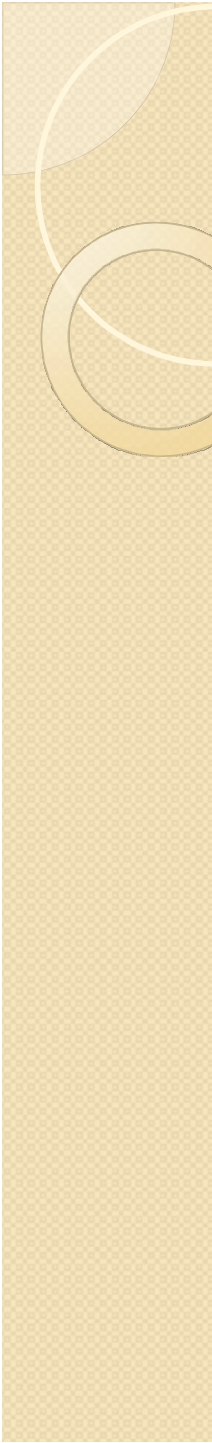


A-

Because someone programmed Java?

No.

It learned from huge datasets containing Java code.



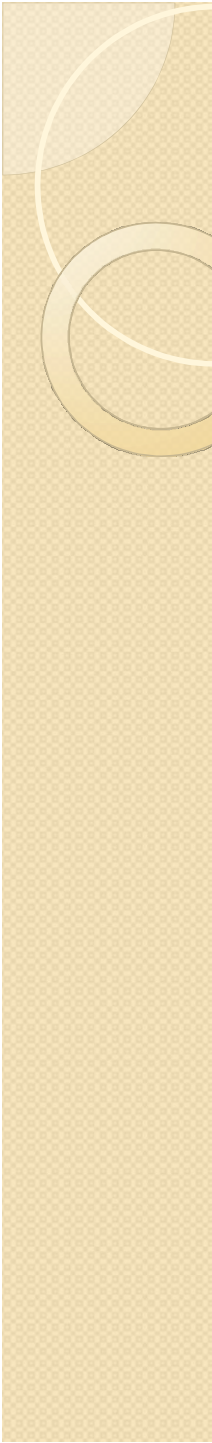
Parameters

Definition

Parameters are the values AI adjusts during training to improve predictions.

In Simple terms

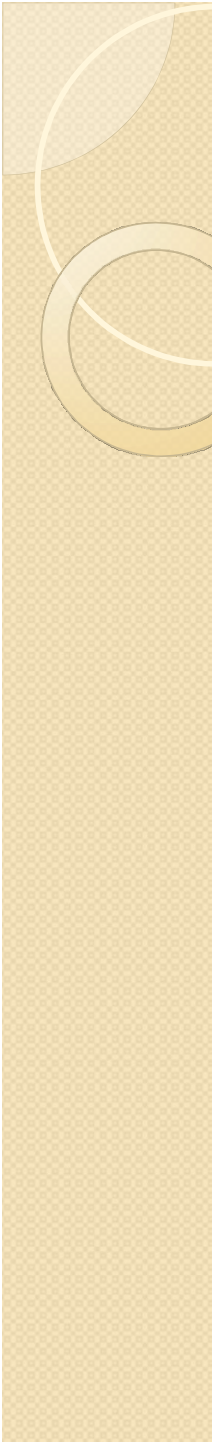
Parameters = AI's memory of what it has learned.



Activity

Q- When you prepare for exams? What stays in your brain?

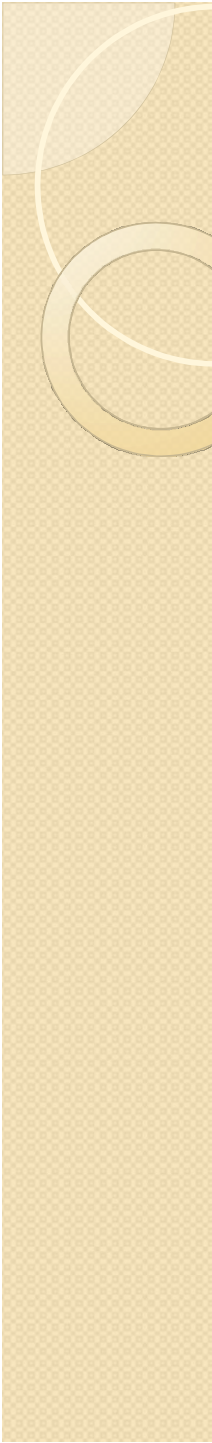
A-



A-
Knowledge.

Similarly,

AI stores learned patterns in parameters.



Small Model vs Large Model

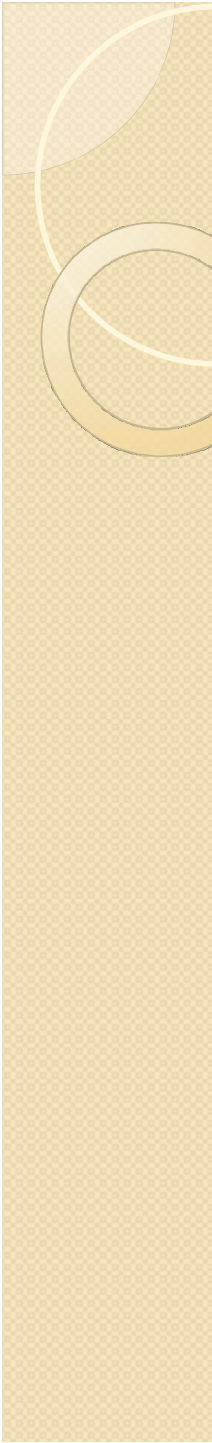
Example-

Small model

Knows basic English.

Large model

- Knows
- Programming
- Medicine
- Law
- Mathematics
- History
- Multiple languages



Activity

Q- Who can answer more questions?

Class 5 student

or

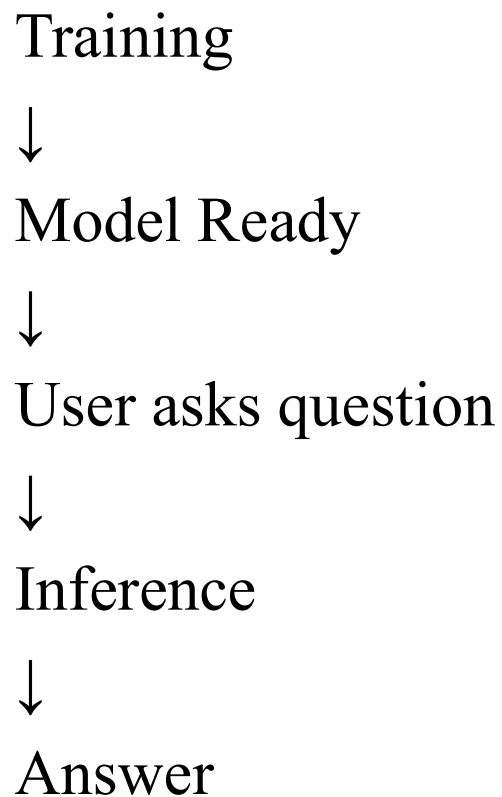
BCA graduate?

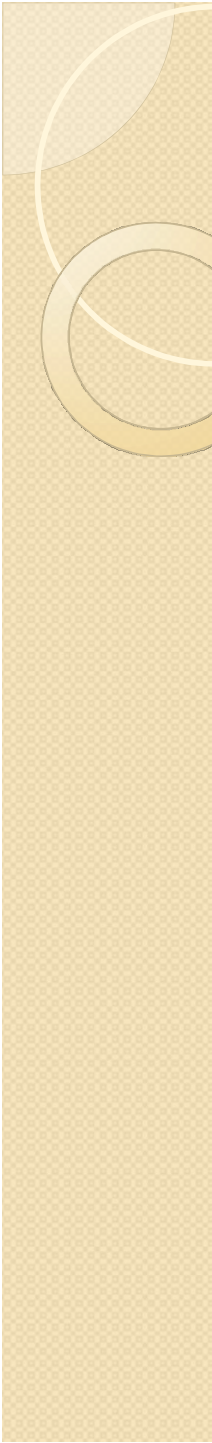
Inference

Definition

Inference is the process of using learned knowledge to answer a new question.

Diagram

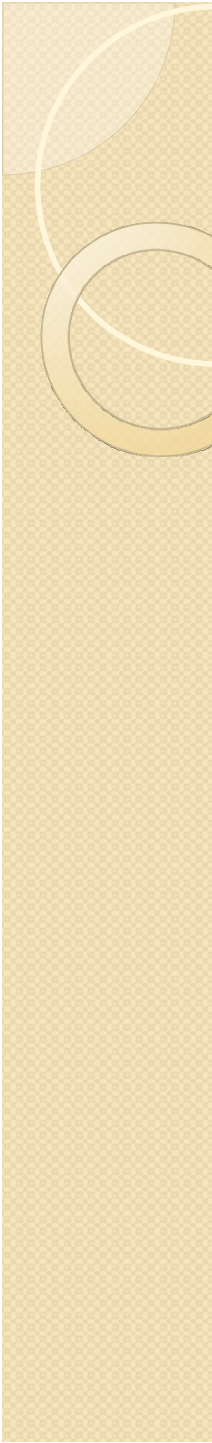




Activity

Q- Try to think training happens once or every question?

A-?



A-

Training:- One huge process.

(example is that feeding a specific model to solve all data structure related coding machine round questions)

Inference:- Every time you ask a specific question

(ask give some specific question on linked list asked in interviews)

Activity

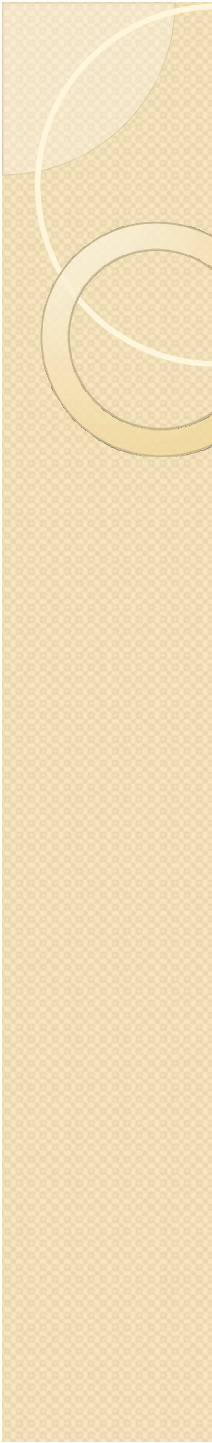
Q- Categorize following activity into training or inference?

A-

Reading textbooks	?
Writing exams	?
Chatgpt answering	?
Teaching a ChatGpt new model	?

Answers:-

Reading textbooks	Training
Writing exams	Inference
Chatgpt answering	Inference
Teaching a ChatGpt new model	Training



Tokens

Definition

AI does not read words. It reads tokens.

Examples

"ChatGPT is amazing"

Break into:-

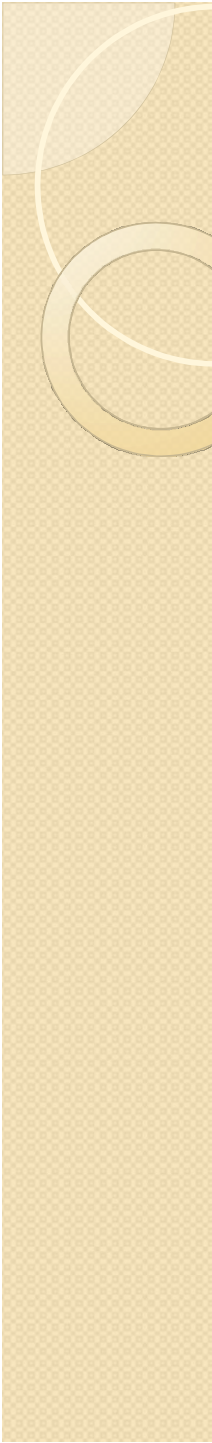
Chat

GPT

is

Amazing

(Not exact tokenization—just for understanding.)

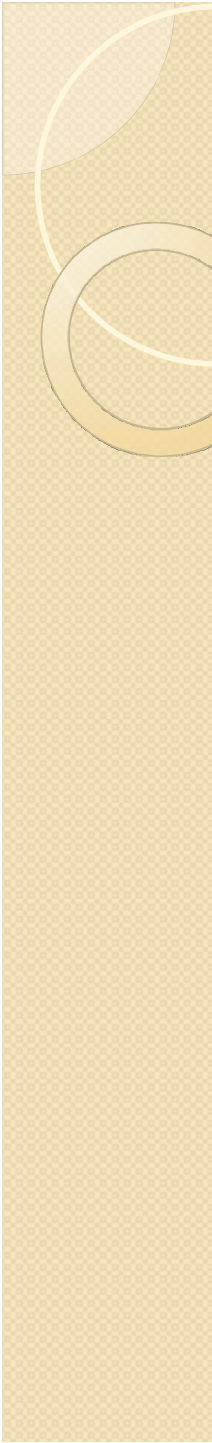


Activity

Q- Split different sentences of your choices into pieces (tokens).

A-

compare with AI tokenization.



Token Prediction

ChatGPT predicts one token at a time.

Live Demo:-

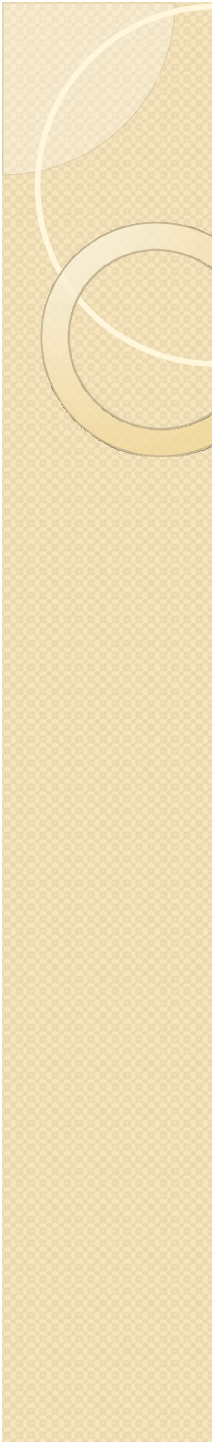
Type:- "The capital of India is" Stop.

Q- What comes next?

A-

Explain

ChatGPT predicts exactly like this.



Activity

Q- Given five incomplete sentences. Predict next word.

The Incomplete Sentences:-

"The dog barked at the..."

"She lost her car..."

"He took a deep..."

"Please turn off the..."

"The sun started to..."

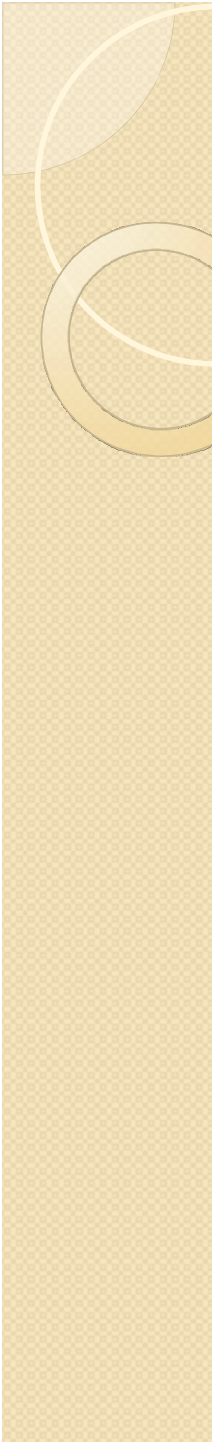
A-

Then ask ChatGPT.

Compare.

Answers

sentences	Student prediction	ChatGpt prediction
The dog barked at the..."	cat / mailman / squirrel	cat / mailman / squirrel
She lost her car..."	keys	keys
He took a deep..."	breath / sigh	breath / sigh
Please turn off the..."	lights / TV / stove	lights
The sun started to..."	set / rise / shine	set / rise / shine



Context Window

Definition

Context Window is the amount of information AI can remember during one conversation.

Example

Conversation

Question 1



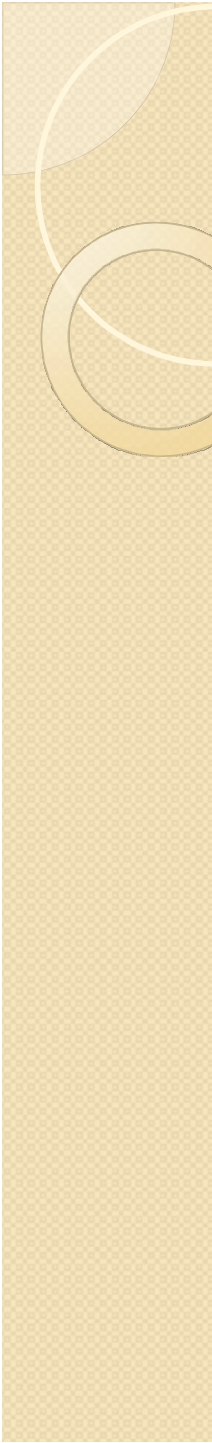
Question 2



Question 3



AI remembers previous questions.



Activity

Memory Game: Hobby chain

Student 1: "I like cricket."

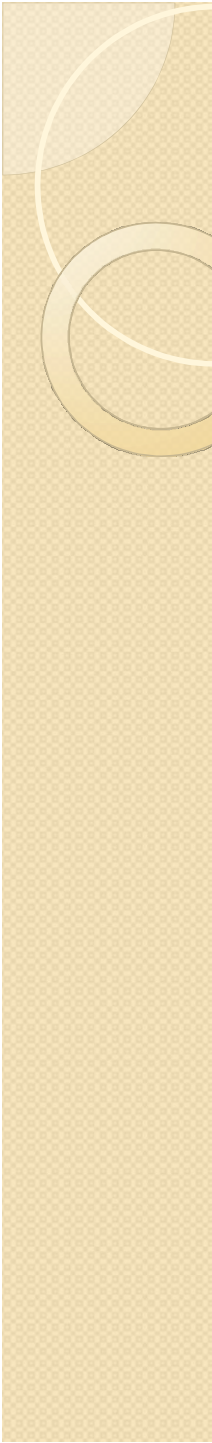
Student 2: "I like cricket, _____."

Student 3: "I like cricket, _____, and _____."

The Breakdown: Eventually, the list gets too long

Humans also have limited memory.

AI has context limits too.

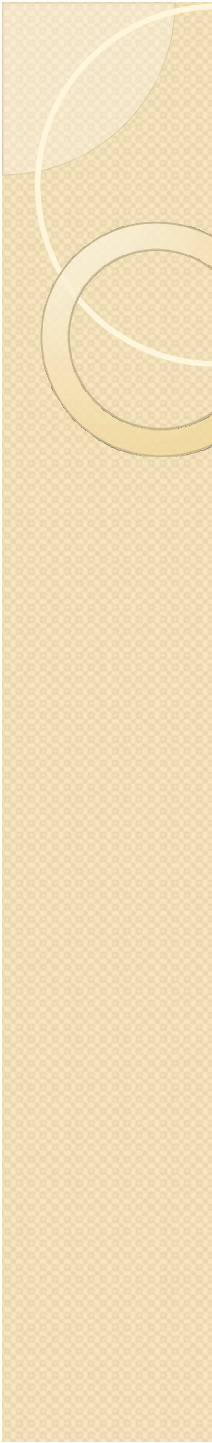


Why ChatGPT Sometimes Forgets

Reasons:-

- Long conversations
- Context limit reached
- New chat
- Incomplete prompt

Q- Has anyone noticed ChatGPT forgetting earlier messages?



Live Demo of chatGpt forgets context

Open ChatGPT

"My name is Rahul."

Then

"What is my name?"

Works.

Now open new chat.

Ask again.

It forgets.

- understand context.

Complete AI Workflow

Data



Dataset



Training



Parameters



Trained Model



Prompt



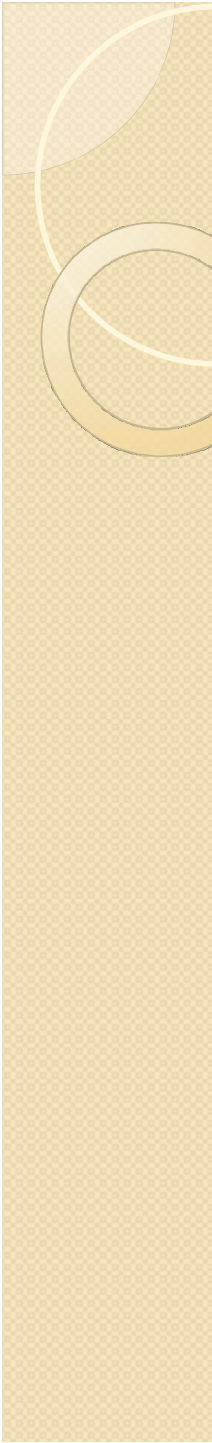
Inference



Token Prediction

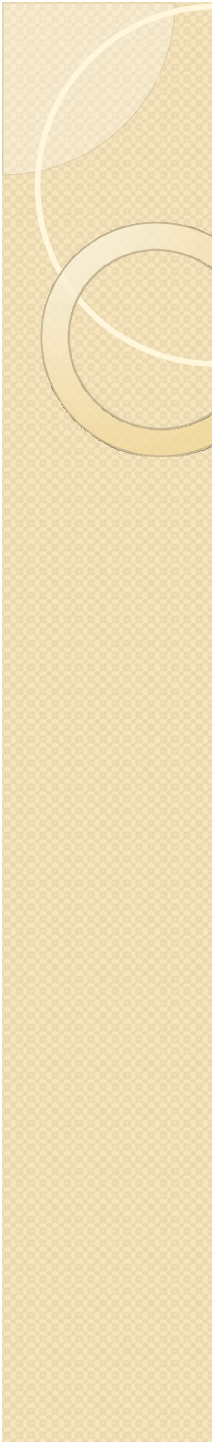


Generated Answer



Activity

Q- explain how complete AI workflow works in your own words.



Class Challenge

Q- Give 5 scenarios. Identify these :

- Dataset
- Training
- Inference
- Tokens
- Context Window

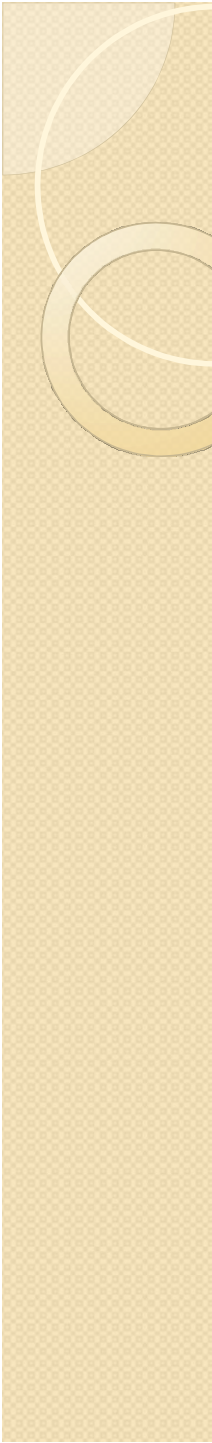
Scenarios:-

- Google Translate
- ChatGPT
- Face Unlock

Scenarios	Dataset	Training	Inference	Tokens	Context window
G translate	Millions of paired sentences translated by humans expert	AI studies these pairs	App instantly generates english version of students text	Individual word or sub-words	Specific paragraph being translated, to ensure words are interpreted accordingly

ChatGpt	Terabytes of public internet texts, books	Model run in supermarket for months	The chatbot generates the functional code on the screen in real-time	Chunks of text or code (roughly 4 characters each) that the model reads and outputs.	The maximum length of the conversation history the chatbot can remember at once.
----------------	--	--	--	--	--

Face unlock	Generic facial databases used by manufacturers,	The phone's neural chip learns the permanent mathematical landmarks of the user's facial structure.	The front camera scans the face in the dark and decides whether to unlock.	Small pixel patches or coordinate coordinates mapped out across the user's eyes, nose, and jaw.	The live frame grabs the facial features and captures the user's face in real time.
--------------------	--	--	---	--	--

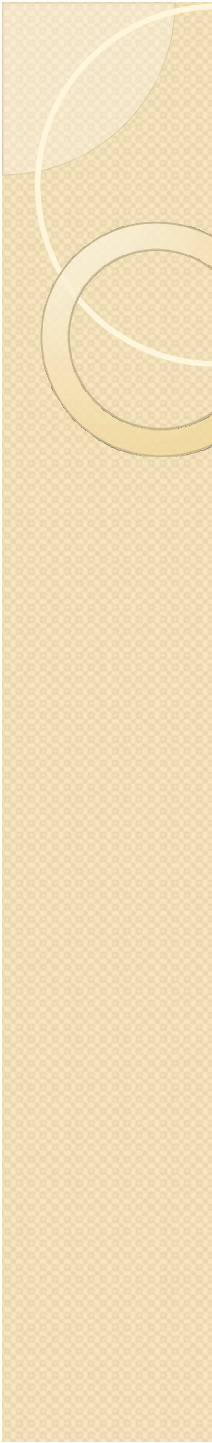


Industry Connection

Q- Where are these concepts used?

- ChatGPT
- Gemini
- Claude
- GitHub Copilot
- Perplexity
- Cursor
- AI Agents

All follow the same basic learning pipeline.



Key Takeaways

- AI learns from datasets.
- Training is learning.
- Inference is answering.
- Parameters store learned knowledge.
- Tokens are the units AI processes.
- Context Window determines how much AI remembers.
- ChatGPT predicts one token at a time.